

The introduction clearly identifies the problem being investigated and distinguishes the manuscript's contribution from conventional logging, governance, auditing, and accountability approaches. The research objective and the four principal contributions are stated clearly at an early stage, and the scope and limitations of the claims are appropriately identified.

The overall design and computational methodology are appropriate to the stated research objectives. The manuscript provides a clear formal criterion and operationalises it using an ontology, established semantic vocabularies, SHACL shapes, SPARQL constraints, and worked validation examples.

One methodological point should, however, be clarified. Definition 1 requires a procedure capable of recovering the truth value of a determination soundly, whereas SHACL primarily establishes whether specified structural elements, types, and relations are present. Structural conformance does not itself establish that the underlying assertions are accurate or sufficient to determine the legally correct truth value. The authors should therefore define SHACL conformance more precisely as a machine-checkable test of structural evidentiary adequacy, rather than directly equating conformance with successful recovery of the determination's truth value.

No additional experiment is necessarily required for the present manuscript, but the reproducibility of the computational implementation should be strengthened by providing a concise repository manifest or reproducibility table identifying the exact artefact versions, files, dependencies, validator configuration, and immutable release corresponding to the manuscript.

The worked validation and comparison between under-typed and typed relational records are clearly presented and directly support the manuscript's principal argument. Tables and figures are generally well integrated and useful for explaining the computational implementation.

There is one consistency issue that should be corrected. Figure 3 refers to five conjunctive properties, whereas Section 3.3 subsequently describes six properties, including intervention duration. The number and definition of these properties should be made consistent across the text, ontology, SHACL/SPARQL implementation, and deposited artefacts.

The discussion is coherent with the preceding analysis and appropriately distinguishes structural evidentiary adequacy from legal compliance, truthfulness, probative force, and successful oversight more generally. The manuscript also relates its contribution to previous work in legal evidence, accountability, semantic technologies, runtime monitoring, and EU AI regulation.

The limitations are clearly acknowledged, particularly the distinction between necessity and sufficiency and the bounded coverage of provenance-capturable events. The conclusion accurately summarizes the contribution and explains how the proposed criterion and executable representation advance the field of computational law.

The references are relevant to the interdisciplinary scope of the manuscript and appropriately cover legal evidence, semantic-web standards, AI accountability, auditing, human oversight, runtime monitoring, and EU AI regulation. Citations are generally well integrated with the claims they support. I did not identify any essential missing citation that would prevent publication.

The level of self-citation appears appropriate in the context of describing and providing access to the deposited ontology and reproducibility artefacts.

The tables and figures are relevant and generally effective in illustrating the formal criterion, SHACL constraints, validation examples, framework comparison, and claim register. They are understandable and appropriately referenced in the manuscript.

One correction is required: Figure 3 describes five conjunctive properties, while the subsequent discussion identifies six. This should be made internally consistent across the figure, text, ontology, and executable artefact.

This is an ambitious, original, and well-structured computational-law contribution. The manuscript identifies an important distinction between merely recording agentic-system activity and producing records capable of supporting legally relevant findings. Its principal contribution—the requirement for legal typing and truth-relevant relations—is clearly formulated, carefully bounded, and operationalised through DPV, PROV-O, SHACL, and SPARQL. The worked examples, claim register, explicit limitations, and proposed empirical evaluation demonstrate strong methodological discipline. The manuscript also succeeds in connecting legal evidence theory, semantic technologies, runtime oversight, and EU AI regulation without claiming that structural conformance automatically establishes legal compliance.

I recommend that the authors address the following three points before publication:

1. Clarify the relationship between Definition 1 and SHACL conformance. Definition 1 requires sound recovery of the truth value of a determination, whereas SHACL primarily verifies the presence of specified structural elements. The manuscript should describe SHACL conformance as a machine-checkable test of structural evidentiary adequacy rather than directly equating conformance with recovery of the legally correct truth value.
2. Correct the inconsistency concerning the intervention properties. Figure 3 refers to five conjunctive properties, while Section 3.3 later lists six, including intervention duration. The prose, ontology, SHACL/SPARQL implementation, and deposited artefact should use the same number and definition of conditions.
3. Strengthen reproducibility documentation. Add a concise reproducibility table or repository manifest identifying the exact files, versions, dependencies, validation tools/configurations, and an immutable release corresponding to the manuscript version.

These are focused and readily addressable revisions and do not require substantial restructuring of the manuscript.